

Macroeconomics Homework Problem

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Part I. Multiple-Choice Questions

MC1. Knowledge Point: GDP and National Income Accounting

Which of the following is included in a country's GDP in the current year?

- (A) The sale of a used car between two households
- (B) The purchase of newly produced machinery by a firm
- (C) The purchase of stocks in the secondary market
- (D) Transfer payments from the government to retirees

MC2. Knowledge Point: Final Goods vs. Intermediate Goods

A bakery buys flour worth 500 yuan and uses it to produce bread sold for 1,200 yuan. How much does this bread production contribute to GDP?

- (A) 500 yuan
- (B) 700 yuan
- (C) 1,200 yuan
- (D) 1,700 yuan

MC3. Knowledge Point: Nominal GDP and Real GDP

If nominal GDP rises by 8% and the GDP deflator rises by 3%, real GDP rises by approximately:

- (A) 3%
- (B) 5%
- (C) 8%
- (D) 11%

MC4. Knowledge Point: CPI

The consumer price index measures:

- (A) the prices of all goods produced domestically

- (B) the prices of goods and services purchased by a representative consumer
- (C) the prices of intermediate goods only
- (D) the unemployment rate

MC5. Knowledge Point: Unemployment Rate

A person who is not working, is available for work, and is actively looking for a job is classified as:

- (A) employed
- (B) unemployed
- (C) out of the labor force
- (D) retired

MC6. Knowledge Point: Types of Unemployment

Unemployment caused by a decline in aggregate demand during a recession is called:

- (A) frictional unemployment
- (B) structural unemployment
- (C) cyclical unemployment
- (D) seasonal unemployment

MC7. Knowledge Point: Inflation

Inflation means:

- (A) all prices rise by the same percentage
- (B) the general price level rises over time
- (C) real GDP falls every year
- (D) money supply necessarily falls

MC8. Knowledge Point: Consumption Function

Suppose the consumption function is $C = 80 + 0.75Y_d$, where Y_d is disposable income. The marginal propensity to consume is:

- (A) 0.25
- (B) 0.75
- (C) 80
- (D) 1.75

MC9. Knowledge Point: Multiplier

If the marginal propensity to consume is 0.8, the simple government expenditure multiplier is:

- (A) 2
- (B) 3
- (C) 4
- (D) 5

MC10. Knowledge Point: Fiscal Policy

Which of the following is an expansionary fiscal policy?

- (A) Raising taxes and reducing government spending
- (B) Lowering taxes or increasing government spending
- (C) Raising reserve requirements
- (D) Selling government bonds by the central bank

MC11. Knowledge Point: Monetary Policy

An open market purchase of government bonds by the central bank will typically:

- (A) reduce the money supply
- (B) increase the money supply
- (C) increase taxes
- (D) reduce government spending

MC12. Knowledge Point: Money Multiplier

If the required reserve ratio is 25%, the simple money multiplier is:

- (A) 2
- (B) 4
- (C) 5
- (D) 8

MC13. Knowledge Point: Functions of Money

Which of the following is *not* a function of money?

- (A) Medium of exchange
- (B) Unit of account
- (C) Store of value
- (D) Determinant of productivity by definition

MC14. Knowledge Point: IS-LM Intuition

A decrease in the interest rate tends to:

- (A) reduce investment spending
- (B) increase investment spending
- (C) reduce consumption automatically to zero
- (D) increase taxes by definition

MC15. Knowledge Point: Aggregate Demand

Other things equal, a fall in the overall price level tends to:

- (A) reduce the quantity of aggregate demand
- (B) increase the quantity of aggregate demand
- (C) shift aggregate demand leftward
- (D) shift long-run aggregate supply leftward

MC16. Knowledge Point: Short-Run Aggregate Supply

In the short run, a rise in firms' production costs tends to:

- (A) shift SRAS to the right
- (B) shift SRAS to the left
- (C) shift AD to the right
- (D) shift LRAS to the right immediately

MC17. Knowledge Point: Economic Growth

In the long run, sustained growth in real GDP per capita is mainly driven by:

- (A) repeated increases in the price level
- (B) growth in productivity and capital accumulation
- (C) cyclical unemployment
- (D) lower labor force participation

MC18. Knowledge Point: Open Economy

Net exports are equal to:

- (A) exports plus imports
- (B) exports minus imports
- (C) consumption minus investment
- (D) taxes minus government spending

MC19. Knowledge Point: Bond Prices and Interest Rates

When market interest rates rise, the price of an existing bond typically:

- (A) rises
- (B) falls
- (C) stays unchanged
- (D) becomes equal to its face value automatically

MC20. Knowledge Point: Financial Markets and Investment

A lower interest rate in financial markets is most likely to:

- (A) discourage firms from borrowing for investment
- (B) encourage firms to borrow and invest
- (C) reduce the present value of future income streams
- (D) reduce the money supply automatically

MC21. Knowledge Point: Stock Market and Household Wealth

If stock prices rise substantially, households may increase consumption because:

- (A) the wealth effect raises desired spending
- (B) unemployment must fall to zero
- (C) taxes automatically decline
- (D) imports necessarily exceed exports

MC22. Knowledge Point: Financial Intermediation

Commercial banks play an important macroeconomic role because they:

- (A) only print money
- (B) channel funds from savers to borrowers
- (C) determine the unemployment rate directly
- (D) set the GDP deflator

MC23. Knowledge Point: Exchange Rate and Net Exports

If a country's currency appreciates, other things equal, its exports tend to:

- (A) rise because domestic goods become cheaper abroad
- (B) fall because domestic goods become more expensive abroad
- (C) remain unchanged by definition
- (D) rise together with imports for the same reason

Part II. Calculation Problems

C1. Knowledge Point: GDP Accounting

Consider an economy with the following data (in billion yuan):

$$C = 500, \quad I = 120, \quad G = 150, \quad X = 80, \quad M = 60.$$

- (1) Calculate GDP using the expenditure approach.
- (2) If depreciation is 40 and net factor income from abroad is 10, calculate GNP and NNP.

C2. Knowledge Point: CPI, Inflation Rate, and Real Wage

A consumer basket contains 2 units of food and 1 unit of clothing. In the base year, food costs 10 and clothing costs 20. In the current year, food costs 12 and clothing costs 25. A worker's nominal wage rises from 100 in the base year to 112 in the current year.

- (1) Calculate the CPI in the base year and in the current year.
- (2) Calculate the inflation rate.
- (3) Has the worker's real wage risen or fallen?

C3. Knowledge Point: Unemployment Rate and Labor Force Participation Rate

In an economy, the adult population is 1,000. Among them, 620 are employed, 80 are unemployed, and the rest are not in the labor force.

- (1) Calculate the labor force.
- (2) Calculate the unemployment rate.
- (3) Calculate the labor force participation rate.

C4. Knowledge Point: Simple Keynesian Model and Government Spending Multiplier

Suppose consumption is

$$C = 100 + 0.8(Y - T),$$

investment is $I = 120$, government spending is $G = 180$, and taxes are $T = 100$.

- (1) Find the equilibrium level of income Y .
- (2) If government spending increases by 20, what is the new equilibrium income?

C5. Knowledge Point: Tax Multiplier

Suppose the marginal propensity to consume is 0.75.

- (1) Find the simple tax multiplier.
- (2) If taxes decrease by 40, by how much does equilibrium income change?

C6. Knowledge Point: Money Creation and Money Multiplier

The required reserve ratio is 10%, and the banking system receives a new deposit of 500.

- (1) What is the simple money multiplier?
- (2) What is the maximum increase in the money supply?

C7. Knowledge Point: Quantity Theory of Money

Suppose the money supply is 1,000, velocity is 4, and real output is 2,000.

- (1) Using the quantity equation, find the price level.
- (2) If money supply rises to 1,200 while velocity and output remain constant, what is the new price level?

C8. Knowledge Point: Financial Markets — Bond Price and Yield

A one-year bond promises to pay 105 yuan next year. If its current price is 100 yuan:

- (1) What is the bond's annual yield?
- (2) If the bond price rises to 102 yuan, what happens to the yield?

C9. Knowledge Point: Financial Markets — Present Value and Investment Decision

A firm can undertake an investment project that costs 100 today and yields 108 next year.

- (1) If the market interest rate is 5%, should the firm invest?
- (2) If the market interest rate rises to 10%, should the firm still invest?

C10. Knowledge Point: Exchange Rate and Net Exports

Suppose a country's exports are 300 and imports are 260.

- (1) Calculate net exports.
- (2) If the domestic currency appreciates and imports rise to 290 while exports fall to 280, what is the new level of net exports?